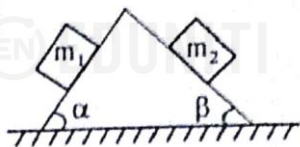
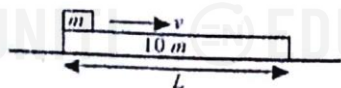


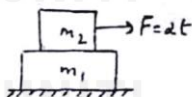
1. Figure shows a smooth wedge placed on a horizontal surface. Two blocks of mass m_1 and m_2 slide on the surface of the wedge. It is observed that wedge does not move. Find the ratio of m_1 and m_2 .



2. A block of mass m is projected on a larger block of mass $10m$ with a velocity v as shown in fig. The larger block is initially at rest and has a length L . The coefficient of friction between the two blocks is μ_2 while that between the lower block and the ground is μ_1 . Given that $\mu_2 > 11\mu_1$, find the minimum value of v such that the mass m falls off the block of mass $10m$.

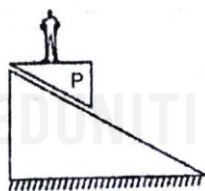


3. In the fig. shown, the coefficient of friction between the blocks is μ and the bottom surface of m_2 is frictionless. Force $F = \alpha t$ is applied on the block of mass m_2 . Here α is a constant and t is the time.



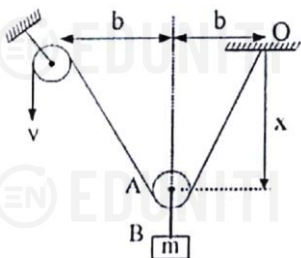
- Find the time t_0 after which the blocks start moving separately.
- Find the acceleration of the blocks as a function of time.
- Draw the acceleration-time graph of both the blocks.

4. A man of mass 80 kg, stands on a horizontal weighing machine, of negligible mass, attached to a massless platform P that slides down at 37° incline. The weighing machine read 72 kg. The man is always at rest w.r.t. weighing machine. Calculate

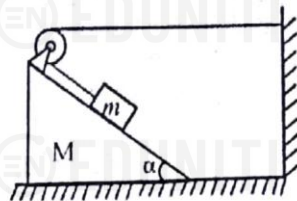


- The vertical acceleration of the man
- The coefficient of kinetic friction μ between the platform and incline.

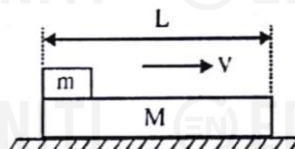
5. The figure shows one end of a string being pulled down by a constant velocity v . Assume that the pulleys are massless and smooth, find the tension in the string AB . The instantaneous position of lower pulley is shown as x .



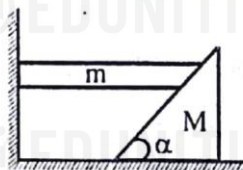
6. In the arrangement shown in figure, the masses m of the block and M of the wedge, as well as angle α , are known. The masses of the pulley and thread are negligible. The friction is absent. Find the acceleration of the wedge M .



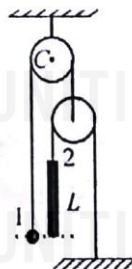
7. Figure shows a small block of mass m kept at the left end of a larger block of mass M and length L . The system can slide on a horizontal surface. The system is started towards right with an initial velocity v . The friction coefficient between the floor and the bigger block is μ and that between two blocks is $\mu/2$. Find the time elapsed before the smaller block separates from the bigger block.



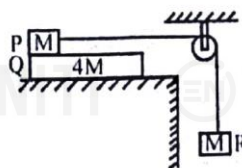
8. A triangular wedge of mass M is placed on a smooth floor near a wall. A prism of mass m is kept as shown in the fig. If the wall is frictionless, find the velocity of the wedge as the prism falls through a distance h .



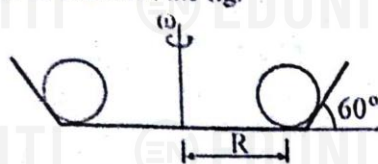
9. In the arrangement shown in figure, the mass of the ball is m . The mass of rod is $M = 2m$ and its length is L . The masses of the pulleys and threads, as well as the friction, are negligible. The ball is set at the same level as the lower end of the rod and then released. How soon will the ball be opposite to the other end of the rod?



10. Three blocks P , Q and R are arranged as shown in the figure. Block Q lies on frictionless surface whereas coefficient of friction between blocks P and Q is 0.25. The string connecting blocks P and R is massless and inextensible, passing over a frictionless massless pulley. The system starts from rest. Find the acceleration of P and Q .



11. Two identical balls each of mass m , rest at opposite ends (of a diameter) of smooth circular pan rotating with angular velocity ω as shown in the fig.

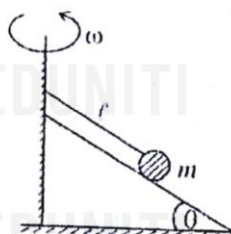


- What is the force by which balls press on the base and side wall of the pan?
- What is the value of ω for which balls lose contact with the bottom surface.

12. A car moves with constant tangential acceleration $a_t = 0.62 \text{ m/s}^2$ along a horizontal surface circumscribing a circle of radius $R = 40 \text{ m}$. The coefficient of static friction between the wheels of the car and surface is $\mu_s = 0.20$. Find the distance covered by the car before it starts slipping, if at initial moment of time, its speed is equal to zero.

13. A small sphere of mass m is tied to the top of the smooth inclined plane with the help of a string of length ℓ . The string and the inclined plane make an angle θ with horizontal. The inclined plane is rotated with a constant angular velocity ω about the vertical axis passing through the end of the string fixed to the plane

- Find the ratio between the tension in the string and the normal reaction between sphere and the plane.
- Find the value of ω at which the sphere will lose contact with the inclined plane.



Answer Key

$$1. \frac{m_1}{m_2} = \frac{\sin 2\beta}{\sin 2\alpha}$$

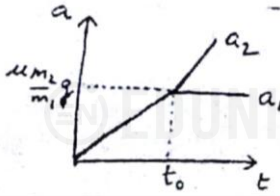
$$2. v > \sqrt{\frac{11(\mu_2 - \mu_1)gL}{5}}$$

$$3. \text{Ans. (a) } t_0 = \frac{\mu m_2 (m_1 + m_2)g}{\alpha m_1}$$

$$(b) \text{ for } t \leq t_0, a_1 = a_2 = \frac{\alpha t}{m_1 + m_2}$$

$$\text{for } t > t_0, a_1 = \frac{\mu m_2}{m_1}g, a_2 = \frac{\alpha}{m_2}t - \mu g$$

(c)



$$4. (a) a_y = 1 \text{ m s}^{-2} \quad (b) \mu = \frac{13}{24}$$

$$5. T = m(g + a) = mg \left(1 + \frac{b^2 v^2}{4gx^3} \right)$$

$$6. A = \frac{mg \sin \alpha}{M + 2m(1 + \cos \alpha)}$$

$$7. t = \sqrt{\frac{2L}{a_r}} = \sqrt{\frac{4ML}{(M+m)\mu g}}$$

$$8. \sqrt{\frac{2gh}{\tan^2 \alpha + M/m}}$$

$$9. t = \sqrt{\frac{10L}{9g}}$$

$$10. a_Q = \mu g/4 = g/16, a_p = g(1-\mu)/2 = 3g/8$$

$$11. (a) N_{\text{side wall}} = \frac{2m\omega^2 R}{\sqrt{3}}, N_{\text{base}} = mg - \frac{m\omega^2 R}{\sqrt{3}}$$

$$(b) \omega = \sqrt{\frac{g\sqrt{3}}{R}}$$

$$12. 60 \text{ m}$$

$$13. (a) \frac{T}{N} = \frac{g \sin \theta + \omega^2 \ell \cos^2 \theta}{g \cos \theta - \omega^2 \ell \cos \theta \sin \theta}$$

$$(b) \omega = \sqrt{\frac{g}{\ell \sin \theta}}$$